

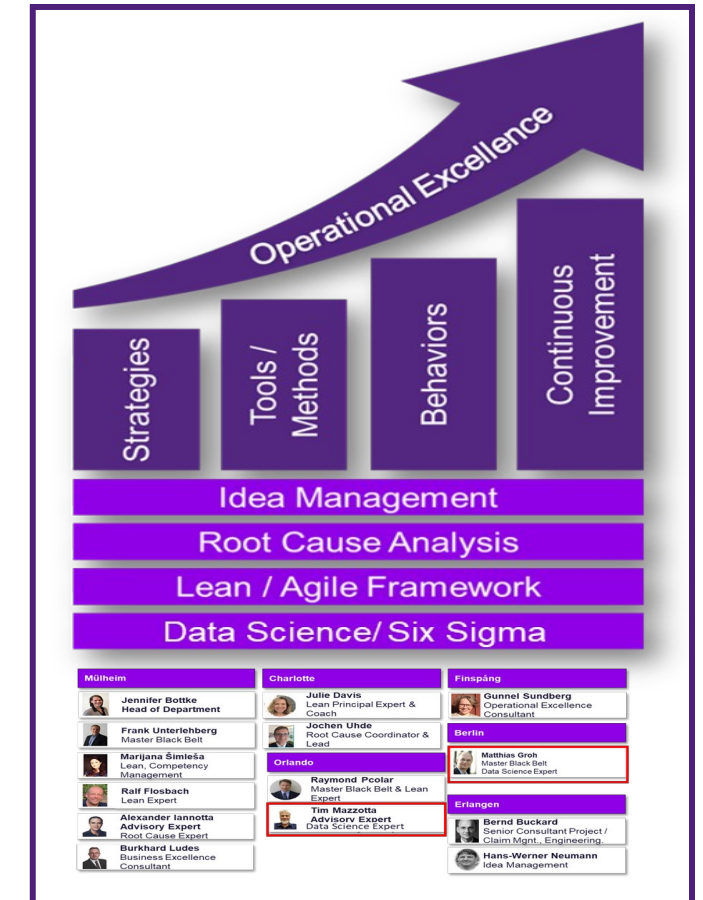
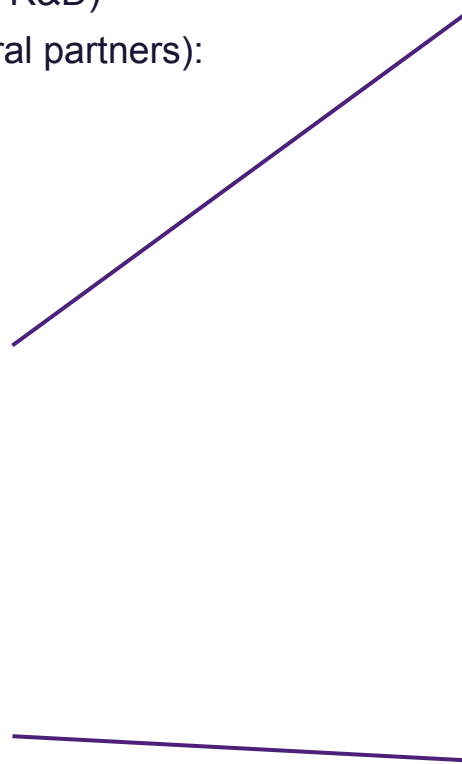
Data-Science Getting Started



Introduction: Data Science @ SE GP G QPI QM P Improvement Projects

We help your employees getting the skills for the digital future:

- Global Data Science Skill Network (with Matt Bryan, LGT R&D)
- Data Science and Machine Learning trainings (with several partners):
 - Currently 11 trainings developed and executed
 - Attended by >500 participants mostly from R&D
- Analytic services in specific improvement projects
- EAI CodeShare (with Thomas Buller, LGN R&D)



Data Science @ QPI QM P:

Matthias Groh
Master Black Belt
Data Science Expert

Tim Mazzotta
Advisory Expert
Data Science Expert

Nicolas Schneider
Coop Master Student

Kay Toepfer
Master Student

We help building the bridge between Data Science and daily engineering business.

Presenters & Content Authors



SE G QPI QM P

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SE GP G SV TI EN RW CRC AC

- Nils Singhof

SE G QPI PI TIM FA

- Ray Witkins

SE G LRE LGN R&D GNS SUP

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- Peter Dumstorff

Goal Data Science Workshops: Basic understanding how to use Python and EnergyAI to access power plant data

Workshop Module	Agenda	Goals	Our approach to teaching:
DS 1 Python Basics	- Jupyter Notebook Intro - Lists, Object Localization - Loops, Dictionaries, If - Definitions - Classes/ Objects/ Methods - Intro to Numpy/ ND-Arrays	Target audience (all modules): - Any engineer who needs to analyze big data - In particular engineers verifying assumptions with power plant data	Handling of questions: - For understanding questions ask right away - For major problems go to separate helpline Teams meeting: one of the teachers will join you as a coach - Expert discussions moved to break or afterwards With a wide range of participant's knowledge: - Target speed is for 20% too fast / for 50% too slow We will record the session (please remind us). Please give us feedback afterwards – we will improve.
DS2 Python: Pandas and Seaborn	<i>Prerequisite Online Training "Transforming Data":</i> <i>Intro data frames + basic editing</i> <i>Overview methods for data frames</i> https://learn.datacamp.com/courses/data-manipulation-with-pandas - Recap - Handling Nans - Resample and Fill - Calculation with dataframes - PickleFormat - Connecting dataframes - Intro Visualization with Seaborn	Prerequisites: - Basic programming knowledge or DS0 - DS1 for DS2, DS2 for DS3 or reading through the scripts New For DS2: "Transforming Data" = free chapter 1 of "Data Manipulation with Pandas" https://learn.datacamp.com/courses/data-manipulation-with-pandas	
DS 3 Data Pulling in Energy AI	- Intro Data Pulling - Time Interval - Plants, Units and Signals - Period, Aggregation - Fill, Thresholds, Poststresholds - Plotting with Seaborn (cont.) - Examples	Teaching goals (all modules): - First steps to learn the topic is taken. - Basic concepts are understood ->just enough to keep learning from online sources. - Everybody learns in the workshop, also trainers. - This is a teaching "Minimal Viable Product"	

How This Training Works

- Experiment, we are all learning
- DS0-> DS1->DS2->DS3 are prerequisites (enough to read through scripts)
- **New For DS2: "Transforming Data" = free chapter 1 of "Data Manipulation with Pandas"** <https://learn.datacamp.com/courses/data-manipulation-with-pandas>
- 1x10 Min Break in each 2h block
- Recording: high demand, few training seats
- Based on interaction
 - Discussions, Exercises + Debriefs
 - Harder in Teams
 - If I should not call on you, please tell us now



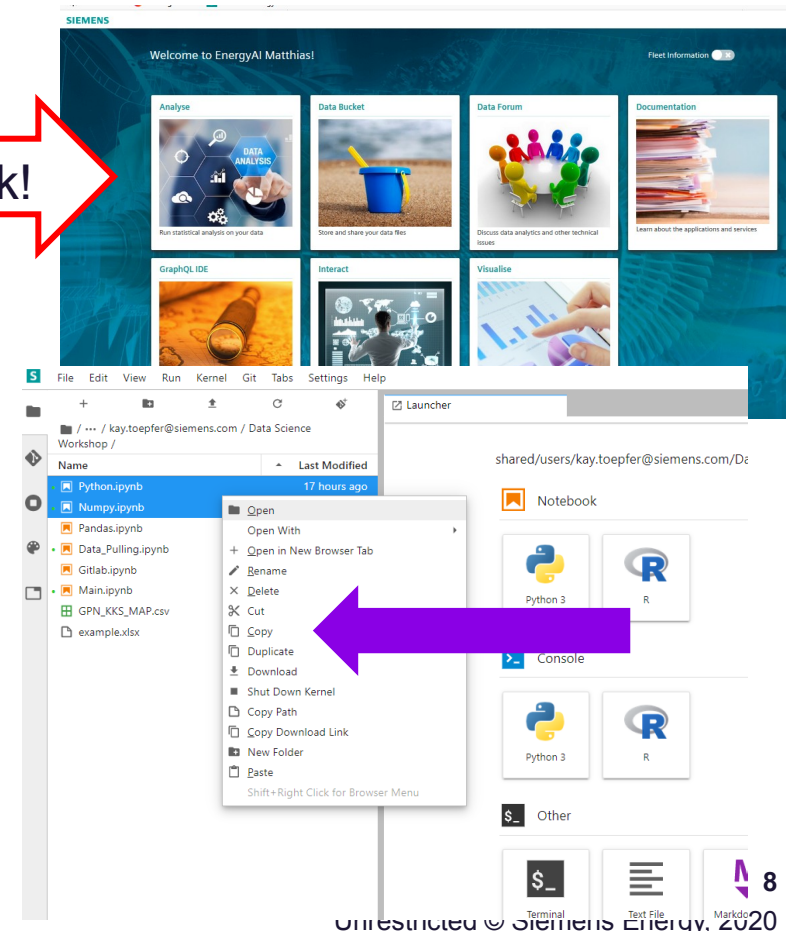
- Our training yammer page:
 - <https://web.yammer.com/main/groups/eyJfdHlwZSI6Ikdyb3VwliwiaWQiOiIxMzAyMjM3MTg0MSJ9/all>
- Evaluation:
 - <https://forms.office.com/Pages/ResponsePage.aspx?id=PqILJW8f80iQ5uJ2ZmS0d-5LdjYRShZEhN3YmsD7St9UREFCNIJTVIRaMVVLNkVOVTNaN09IMTA3MC4u>
 -

Introduction

1. Name, Department
2. All prerequisites fulfilled? EnergyAI, knowledge of previous trainings (DS0->DS1->DS2->DS3)
3. Preexisting knowledge program, data science, statistics, Python, R, ...
4. Expectations

Side Task:

- Go to Energy AI <https://energyai.siemens.cloud/home> and clicked on Analyze.
- Please use Edge, Firefox or Chrome, not MS Explorer
- Copy all Files from shared->User->Kay.Toepfer...->DataScienceWorkshop
- Follow detailed instructions starting on slide 25, if this does not work go to the helpline



DRAFT: Data Structures in Python (Grey), Numpy (Green) and Pandas (Orange)

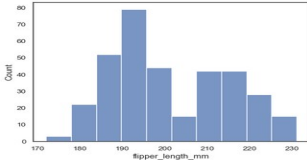
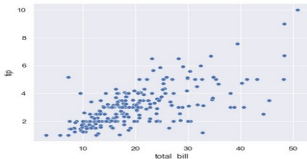
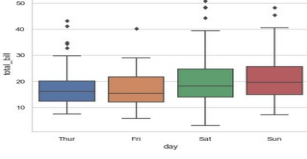
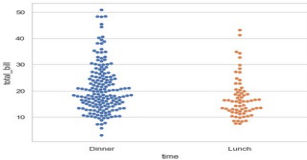
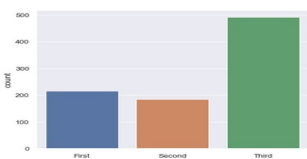
	Description	Useful for...	Example
Lists	Store collection of heterogeneous items – immutable and mutable objects	- Simple listing - Special programming techniques (Stacks, Queue, Graphs, Trees)	[item, item, item]
Dictionaries	- Collection of key-value pairs - Key: immutable objects - Value: heterogeneous items – immutable and mutable objects	Creating loops to do the same operation on many dataset	{key1:item1, key2:item2}
Files	Store and retrieve previously stored information		.csv, .hdf5, .xlsx
Tuples	Tuples are data structures in which the content can not be changed after creation (no deletion, add or edit)	Prevents data manipulation	(item, item, item)
Sets	Collection of unique objects	- Creating lists that only hold unique values - Helpful when going through huge dataset	{item, item, item}
Numpy ND-Arrays	Store collection of data of the same type	- Dealing with large collection of homogeneous data types: easier to use, faster and uses lesser memory than lists - Support vectorized operations - Work efficiently with large datasets with lots of empty cells	<pre>[[1. 1. 2.] [3. 5. 8.] [5. 3. 2.]]</pre>
Dataframes	2-dimensional labeled data structure - Columns don't have to have the same data type	- Data mining / manipulation - Labeling - Multiindexing	Like this table
Series	One-dimensional labeled array capable of holding any data types	Axis labels	<pre>a 1 b 2 c 3</pre>

Source: www.datacamp.com, pandas.pydata.org; Also see: <https://www.datacamp.com/community/tutorials/data-structures-python>

DRAFT: A Selection of Important Libraries in Python

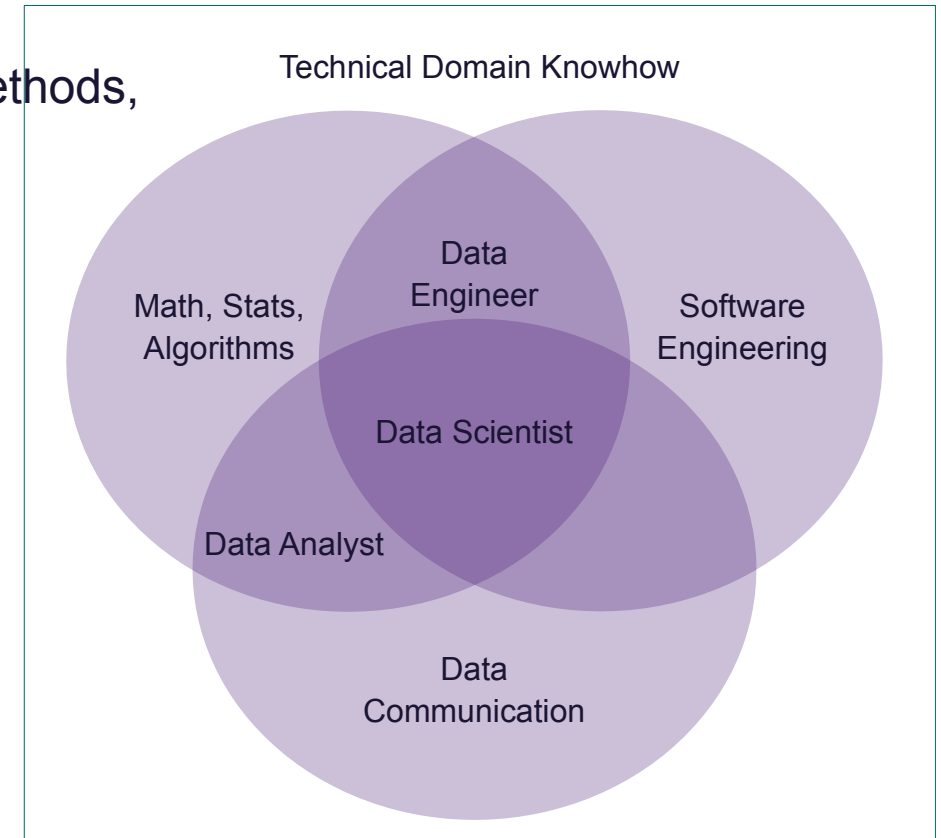
Category	Name	Description	Link
Data Handling Libraries	Numpy	Basic Library for scientific computing in Python (linear algebra, numerical functionalities etc.)	https://numpy.org/
	Pandas	Package especially built for data analysis	https://pandas.pydata.org/
	Scipy	Mathematical and engineering functions (e.g. optimization and fits, numerical integration...)	https://www.scipy.org/
Plotting Libraries	Matplotlib	2D plotting library for basic plotting	https://matplotlib.org/
	Seaborn	Is built on top of matplotlib, thus offers more functionalities	https://seaborn.pydata.org/
	Plotly	Is built for interactive graphs	https://plot.ly/python/
Machine Learning Libraries	TensorFlow	For machine learning applications (e.g. voice recognition)	https://www.tensorflow.org/
	SciKit-Learn	For machine learning applications (e.g. regression models, clustering, vector-machines...)	https://scikit-learn.org/stable/
Other Useful Libraries	Time	Provides all time-related functions	https://docs.python.org/3/library/time.html
	Sys	For interaction with interpreter (advanced)	https://docs.python.org/3/library/sys.html

Overview: Essential Graphs of Seaborn

Name	In SNS	Useful for	Plot	Relative to ...	Target variable
Histogram	.histplot()	- Central tendency - Spread - Distribution - Outliers			Continuous data
Scatterplot	.relplot() X-Axis is continuous	- Correlation - Outliers		... other continuous data	
Lineplot		- Time effect - Spread - Outliers		... time	
Boxplot	.catplot() X-Axis is categorical	- Group effects - Central tendency - Spread - Distribution - Outliers		... categorical data	Quantitative data
Swarmplot					
Countplot		Counts group effects		... categorical data	

Definition Data Science

Data science is a multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data. ^[1]



Data science includes “new” (Machine Learning) and “old” (conventional analytics).

[1] en.wikipedia.org/wiki/Data_science

Data Science for Field Issues (1 of 2)

The Data Request Helper

https://wse10.siemens-energy.com/content/P0002338/_layouts/15/xlviewer.aspx?id=/content/P0002338/Documents/Data_Request_Helper.xlsx

Data Request Helper												
Last updated on: 26.09.2019 This is a living document. If you have updates, changes or requests, please contact grahm.muth@siemens.com												
A field issue / RCA / task force manager needs data to verify a technical hypothesis:	Count and Compare Starts, OH, Trips etc.			Go-to People (let others do it)								
	Plot Chart of one Sensor			Tools (do it yourself)								
	Analyze several Sensors and Events Simultaneously			All tools include at least PDC scope data								
	Engine Configuration and Parts Data			All tools include at least PDC scope data								
A field issue / RCA / task force manager needs data to verify a technical hypothesis:	Count and Compare Starts, OH, Trips etc.			Fleet Data Management (GP SPG TI PA FDM CID / ROW): fdm_analytics.energy@siemens.com Product Analytics (GP SPG TI PA AS): Thomas Hahner http://intranet.for.siemens.com/cma/010/en/about/org/Documents/Go-to/20People.xlsx								
	Plot Chart of one Sensor			Product Analytics (GP SPG TI PA AS): Thomas Hahner Operational Excellence (GP PGO PR QM PPR OE): Jennifer Bottke								
	Analyze several Sensors and Events Simultaneously			Product Analytics (GP SPG TI PA AS): Thomas Hahner Proababilities (GP PGO TI TEC PRD) Operational Excellence (GP PGO PR QM PPR OE): Jennifer Bottke								
	Engine Configuration and Parts Data			Proababilities (GP PGO TI TEC PRD) Fleet Data Management (GP SPG TI PA FDM CID / ROW): fdm_analytics.energy@siemens.com Additional contact people will follow								
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Why EnergyAI?



The EnergyAI - Analysis platform...

- ...offers an efficient interface to handle large data requests on multiple plants or units
- ...enables the user to perform postprocessing on-the-fly, without further data handling steps
- ...provides a standardized GPN System to address signals across all plants and units
- ...executes all operations outsourced on a server and makes results available for client

Additional Features in EnergyAI

Power and Gas – Large Gas Turbines, Generators



Advanced Settings: (1 of 2) to Easier Move between Folders for Opened Notebooks

1) Click: Settings

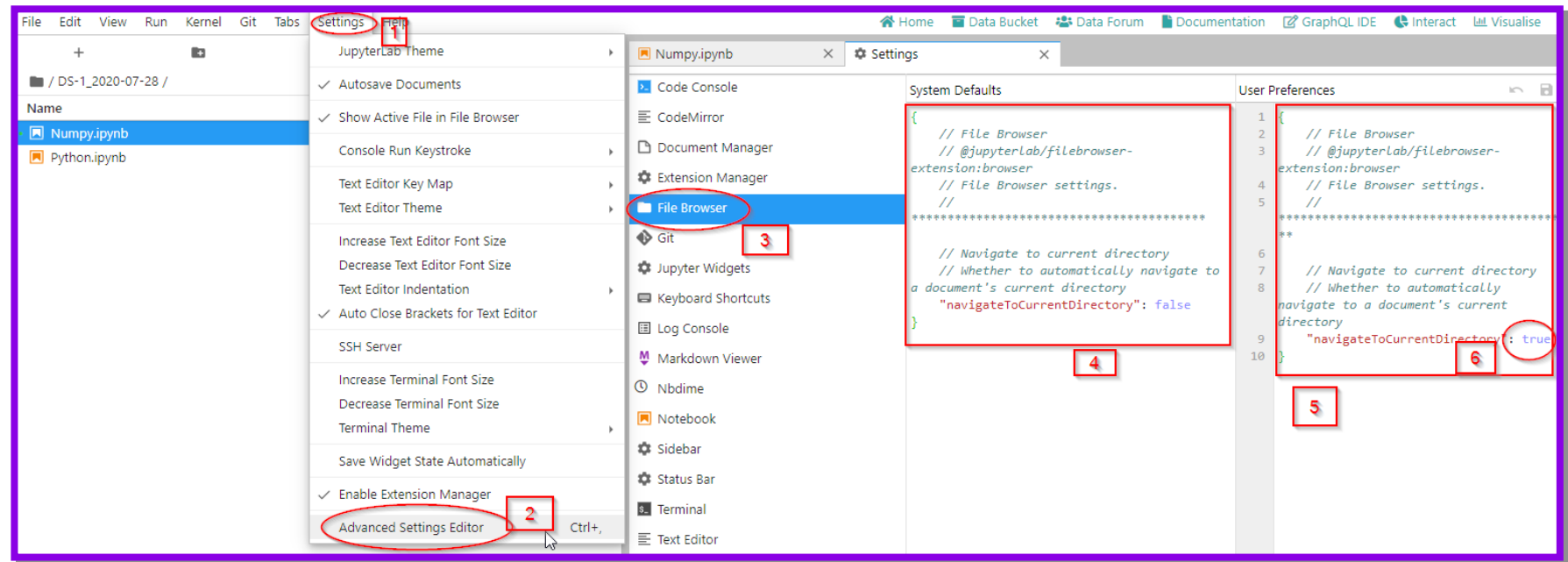
2) Click: Advanced
Settings Editor

3) Click: File Browser

4) Copy

5) Paste

6) Modify



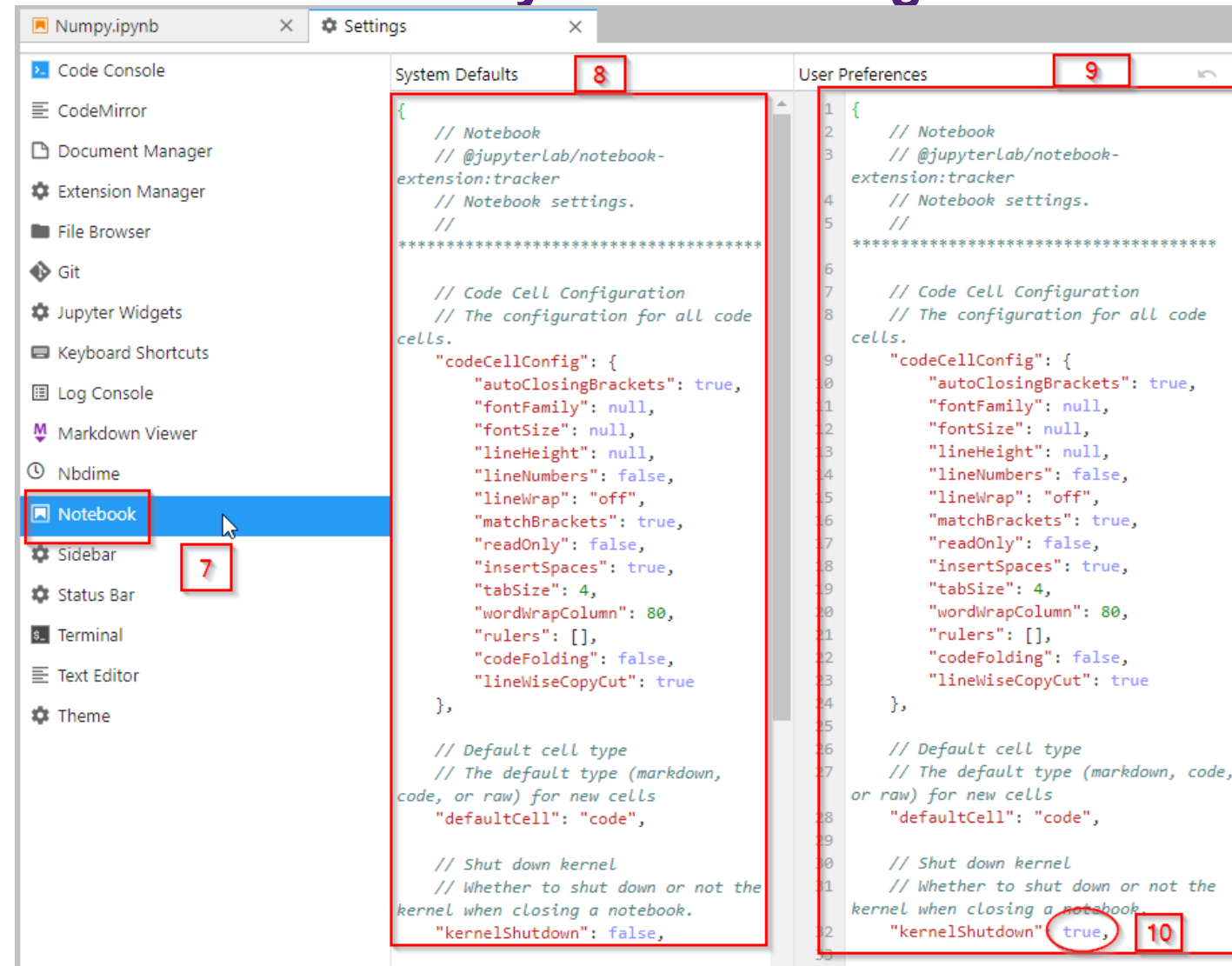
Advanced Settings: (2 of 2) to Shut Down Kernel Automatically when Closing Notebook

7) Click: Notebook

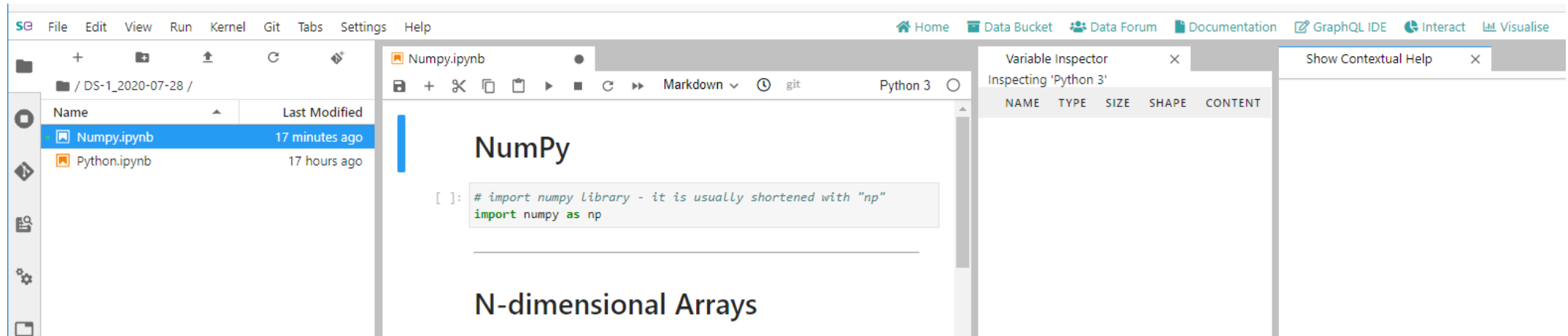
8) Copy

9) Paste

10) Modify

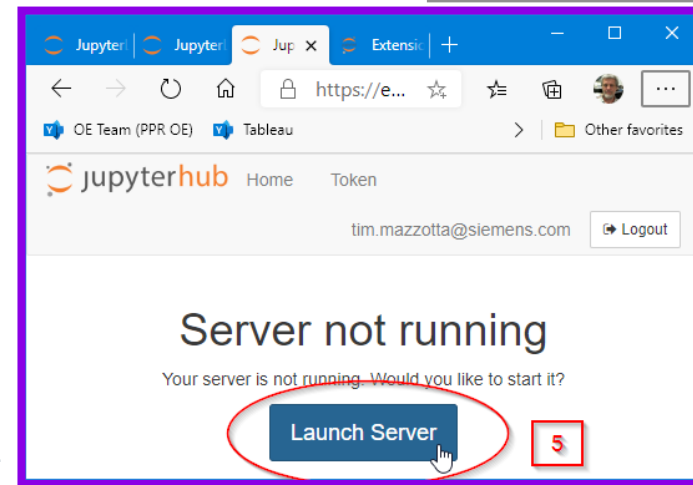
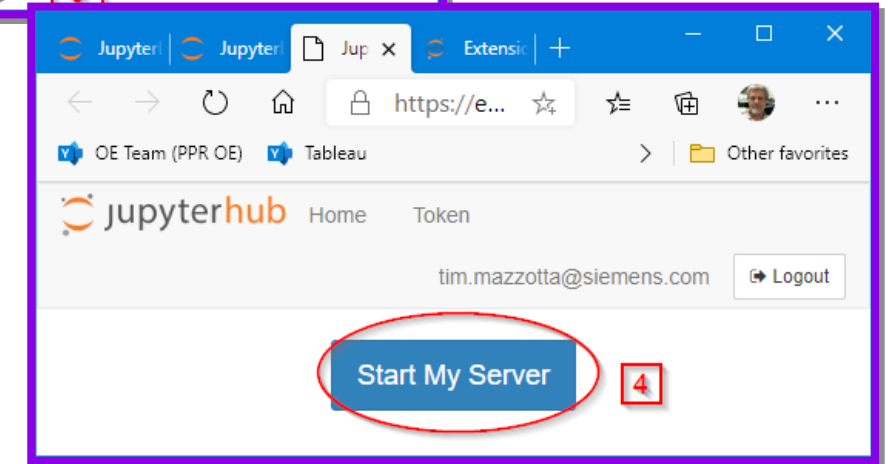
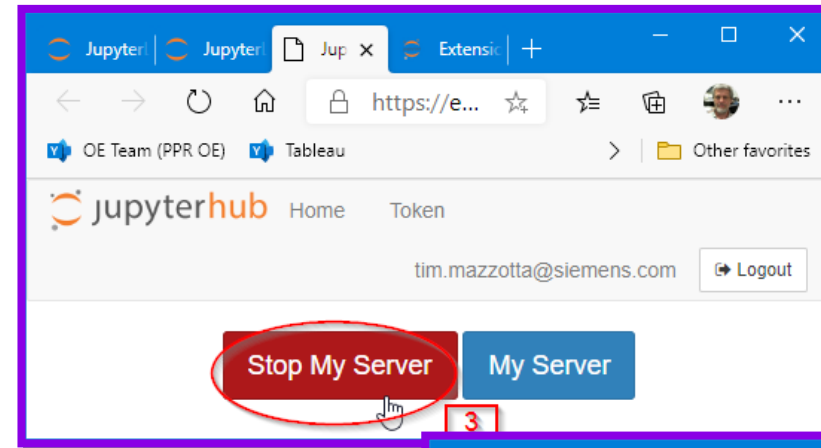
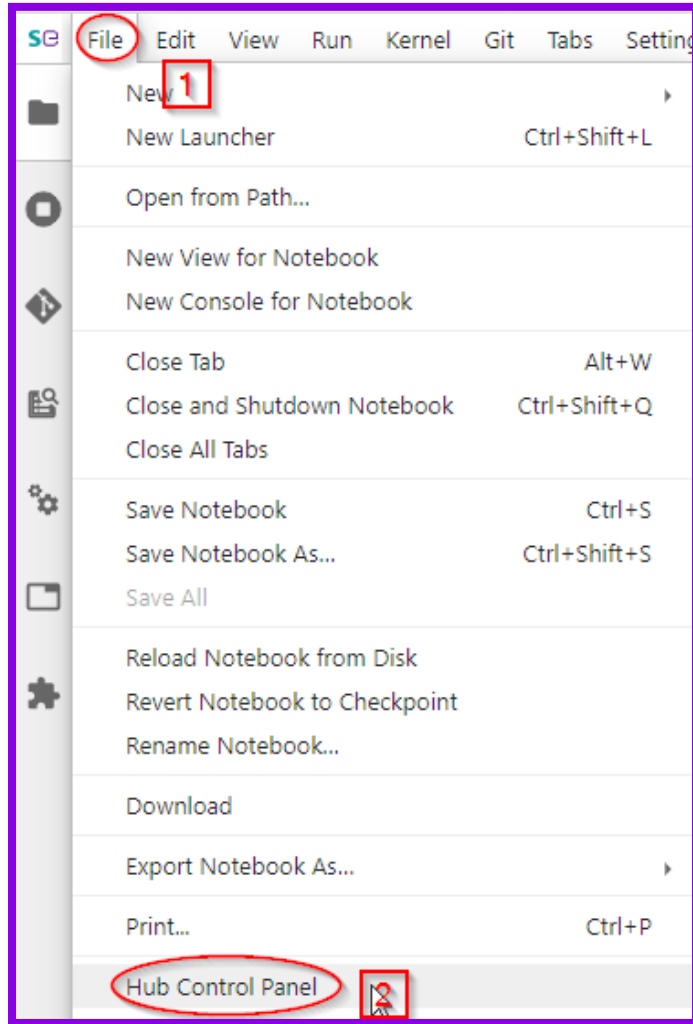


Open up a file and set up view by docking Context Help and Variable Inspector



Trouble Shooting !!! ONLY RUN IF PROBLEMS !!!!

1) info



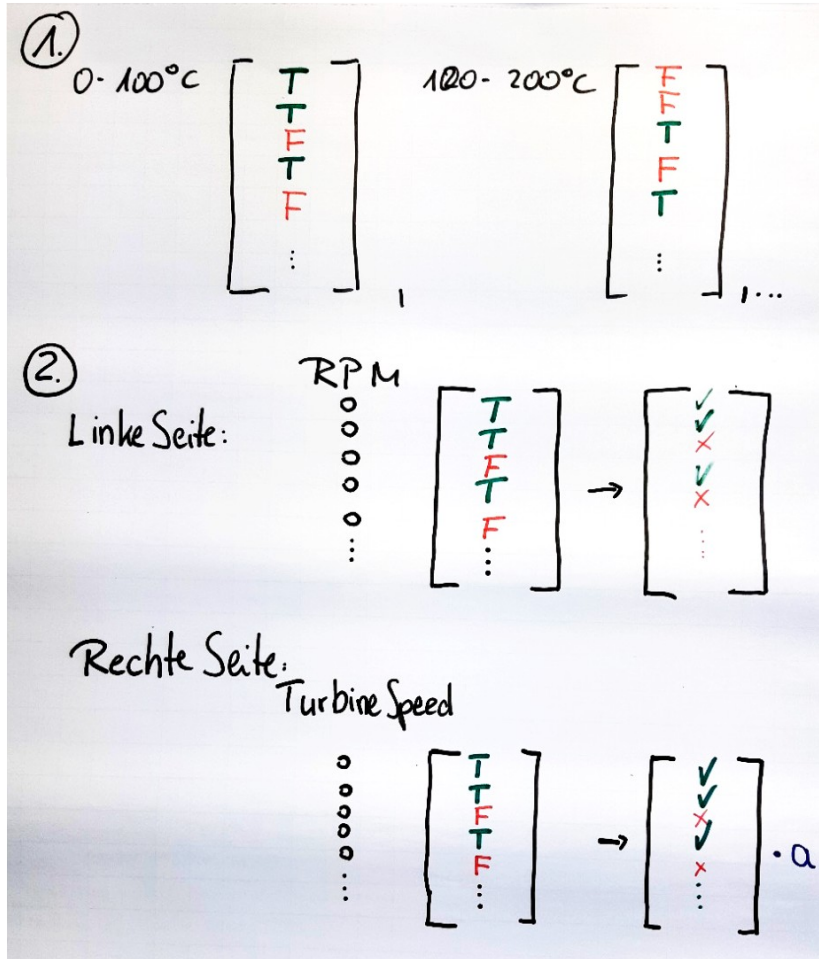
Additional Slides for Data Science 2

Vectorized Calculations

Power and Gas – Large Gas Turbines, Generators



How to Calculate Efficiently in Pandas



Temp zw. 0-100°C TS · 60 · 0,8

- -	100 - 200°C	- -	· 0,85
- -	200 - 300°C	- -	· 0,9
- -	300 - 400°C	- -	· 0,95
- -	400 - 500°C	- -	· 0,99

Turbine Speed · 60 · a

abh. von Temp.

Additional Slides for Data Science 3

Power and Gas – Large Gas Turbines, Generators



Fill

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Forward Fill

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Box-Plot is a plot to show most things a Histogram shows by using much less space and therefore easy allows to compare many distributions. However, it is a bit harder to read than other graphs.

